

4 CAD Laboratory Session n°4 - Design of a low-voltage band-gap voltage reference

This lab session focuses on the design, simulation, and optimization of a Low-Voltage Bandgap Voltage Reference.

4.1 Temperature dependence of forward diode voltage

The schematic used for this part of the laboratory is shown in Fig. 39. The diode used (DIOP_L130E) is a p^+n diode where its saturation current is as follows :

$$I_S \cong A_D \cdot \frac{q n_i^2 D_p}{N_{NW} W_{NW}}$$

where A_D is the diode area, $D_p = \frac{k_B T}{q} \mu_p$ is the hole diffusion coefficient in the n-well, N_{NW} is the n-well doping level, W_{NW} is the n-well effective size

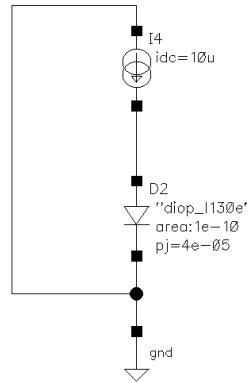


Figure 39: Forward Diode Voltage

When the diode is forward-biased, the direct current I_D is related to the diode voltage V_D by the following equation :

$$I_D = I_S \cdot \left[\exp\left(\frac{V_D}{U_T}\right) \right] \cong I_S \cdot \exp\left(\frac{V_D}{U_T}\right) \Rightarrow V_D = U_T \cdot \ln\left(\frac{I_D}{I_S}\right)$$

$$I_S \cong A_D \cdot \frac{q n_i^2 D_p}{N_{NW} W_{NW}} = I_{S0} \exp\left(-\frac{E_{G0}}{k_B T}\right) = I_{S0} \exp\left(-\frac{V_{G0}}{U_T}\right), \quad \text{where : } V_{G0} = \frac{E_{G0}}{q}$$

E_{G0} is the energy gap of Si at $T = 0K$, and I_{S0} is a coefficient with a negligible dependence on T with respect to the exponential factor. Then :

$$V_D = U_T \cdot \ln\left(\frac{I_D}{I_{S0}}\right) + V_{G0} = -U_T \cdot \ln\left(\frac{I_{S0}}{I_D}\right) + V_{G0}, \quad \text{where : } V_{G0} = 1.205 \text{ V}$$

Given that $I_{S0} > I_D$, then V_D temperature dependence is negative:

$$\frac{\partial V_D}{\partial T}(T) = -\frac{k_B}{q} \cdot \ln\left(\frac{I_{S0}}{I_D}\right) = -\frac{V_{G0} - V_D}{T}$$

that is, a Complementary To Absolute Temperature (CTAT) behavior.

4.2 LV-BGR

4.2.1 LV-BGR Schematic

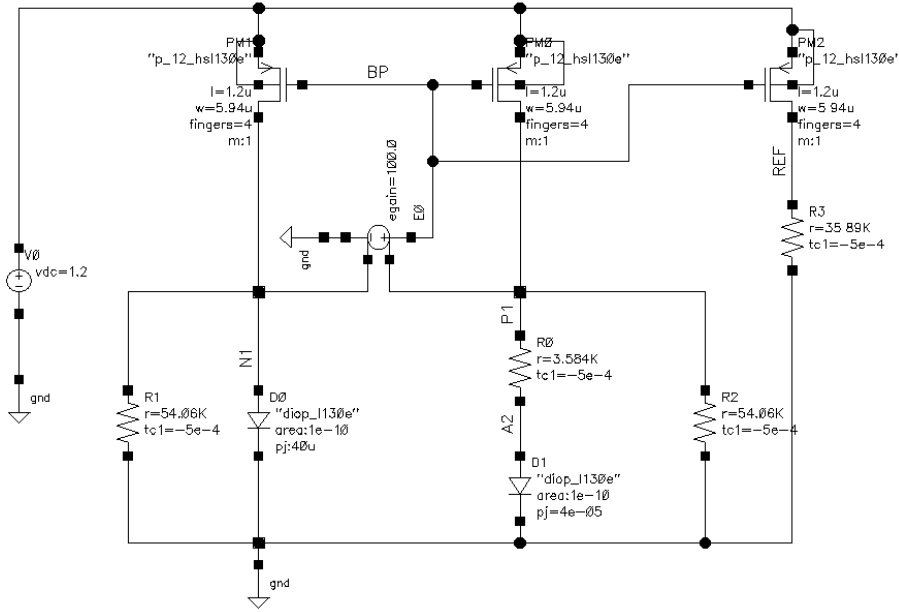


Figure 40: LV-BGR Schematic

4.2.2 Desing of the LV-BGR

A brief summary of the equation that will be used later on are displayed below :

$$I_1 = \frac{k_B T}{q} \cdot \frac{\ln(M)}{R_1} \quad V_{D1} = \frac{k_B T}{q} \cdot \ln\left(\frac{I_1}{I_S}\right)$$

$$I_2 = \frac{V_{D1}}{R_2}$$

$$V_{REF} = R_3 (I_1 + I_2) = \frac{R_3}{R_2} \left[V_{D1} + \frac{k_B T}{q} \cdot \frac{R_2}{R_1} \ln(M) \right]$$

$$V_{D1} \cong V_{G0} - \frac{k_B T}{q} \ln\left(\frac{I_{S0}}{I_1}\right)$$

Since
$$\frac{\partial V_{REF}}{\partial T} \cong -\frac{k_B}{q} \ln\left(\frac{I_{S0}}{I_1}\right) + \frac{k_B}{q} \cdot \frac{R_2}{R_1} \ln(M)$$

then, by setting

$$\frac{R_2}{R_1} \ln(M) = \ln\left(\frac{I_{S0}}{I_1}\right) \implies \frac{\partial V_{REF}}{\partial T} \cong 0 \quad \text{and} \quad V_{REF} \cong \frac{R_3}{R_2} V_{G0}$$

The specifics for the LV-BGR are the following :

- $V_{REF} = 0.8 \pm 0.1\%$
- $TC_{REF} \leq 2 \cdot 10^{-6} K^{-1}$ at $T = 27C$
- $T = -40C \div 85C$
- $V_{DD} = 1.2 V \pm 10\%$

Some parameters are known :

$$I_S = 6.896 \cdot 10^{-17} A \implies I_{S0} = 1.207 \cdot 10^4 A$$

$$V_{G0} = 1.205 V$$

Initially, there were imposed the following parameters :

$$I_1 = 10 \mu A \quad M = 4$$

From the first equation in the summary it was possible to evaluate the resistance R_1 :

$$R_1 = \frac{k_B T}{q} \cdot \frac{\ln(M)}{I_1} = 3.584 k\Omega$$

while the forward voltage of D1 is :

$$V_{D1} = \frac{k_B T}{q} \cdot \ln\left(\frac{I_1}{I_S}\right) \approx 0.6644 V$$

Thanks to the fifth and sixth equation in the summary, R_2 comes out to be :

$$R_2 = \frac{\ln\left(\frac{I_{S0}}{I_1}\right)}{\ln(M)} \cdot R_1 = 54.06 k\Omega$$

$$I_2 = \frac{V_{D1}}{R_2} = 12.29 \mu A$$

Using the last equation in the summary, R_3 was obtained

$$R_3 = \frac{V_{REF}}{V_{G0}} \cdot R_2 = 35.89 k\Omega$$

After obtaining these values for the resistances and the currents, it was possible to size the p-type transistor of the current mirror, knowing that $I_D = I_1 + I_2 = 22.29 \mu A$ and imposing the gate length of transistor $M3$, $M4$ and $M5$ equal to $L = 1.2 \mu m$ and the overdrive voltage $|V_{OV}| = 0.3 V$

$$W_5 = W_4 = W_3 = \frac{2 I_D L}{k'_p V_{OV}^2} = 5.994 \mu m$$

4.2.3 LV-BGR Characterization

To account for slight non-linearities in the curves, the following temperature coefficients were extracted locally around 27°C using closely spaced markers. The analysis was conducted via a DC temperature sweep ranging from 40°C to 85°C .

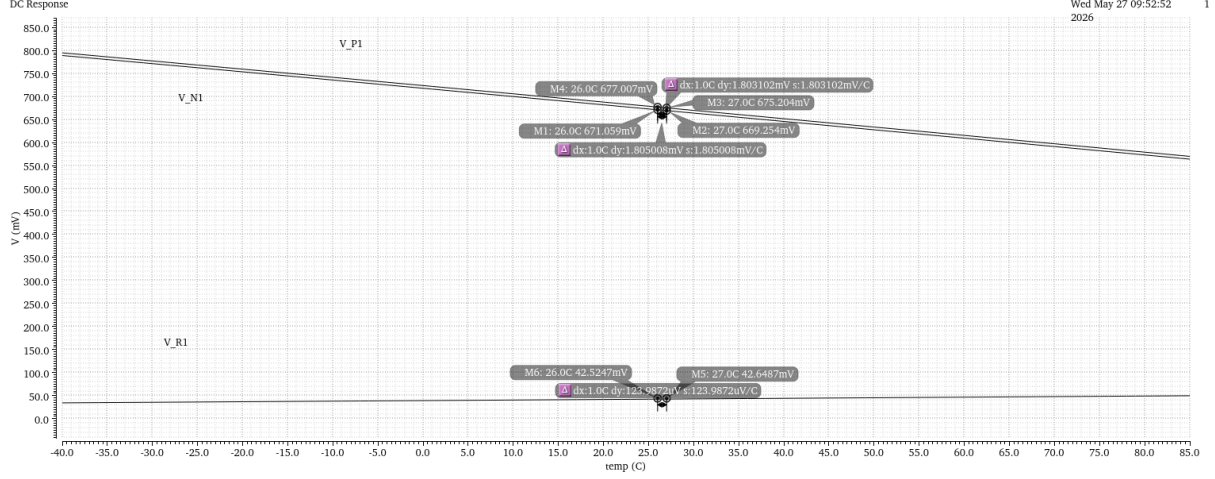


Figure 41: Voltages V_{N1} V_{P1} and the resistance R_1

As it is possible to see in Fig. 41, the voltages V_{N1} and V_{P1} display the expected CTAT behavior:

$$\frac{\partial V_D}{\partial T}(T) \cong -1.8 \frac{\text{mV}}{\text{C}}$$

while the voltage drop across R_1 shows PTAT behavior

$$\frac{\partial V_{R1}}{\partial T} = 123.987 \frac{\mu\text{V}}{\text{C}}$$

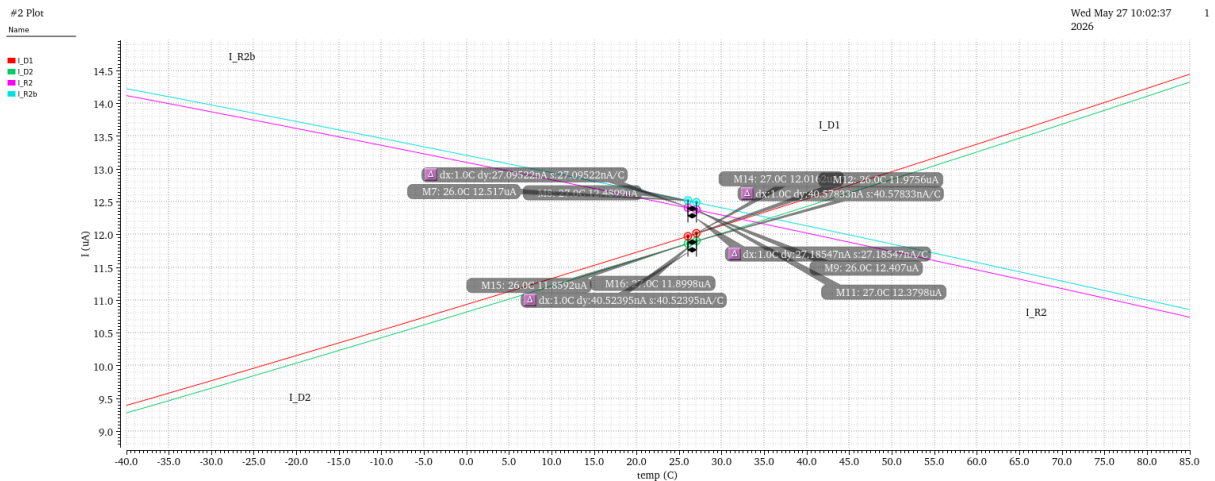


Figure 42: Voltages V_{N1} V_{P1} and the resistance R_1

Regarding the currents flowing through the resistor R_2 and the diode currents I_{D1} and I_{D2} , thanks to Fig. 42 it is possible to confirm that I_{D1} and I_{D2} are almost the same, but

the current flowing in the first diode is slightly above $10 \mu A$ as it was stated. The slope for the diode currents and R_2 are given below :

$$\frac{\partial I_D}{\partial T} \approx 40.5 \frac{nA}{C} \quad \frac{\partial I_{R2}}{\partial T} \approx 27.1 \frac{nA}{C}$$

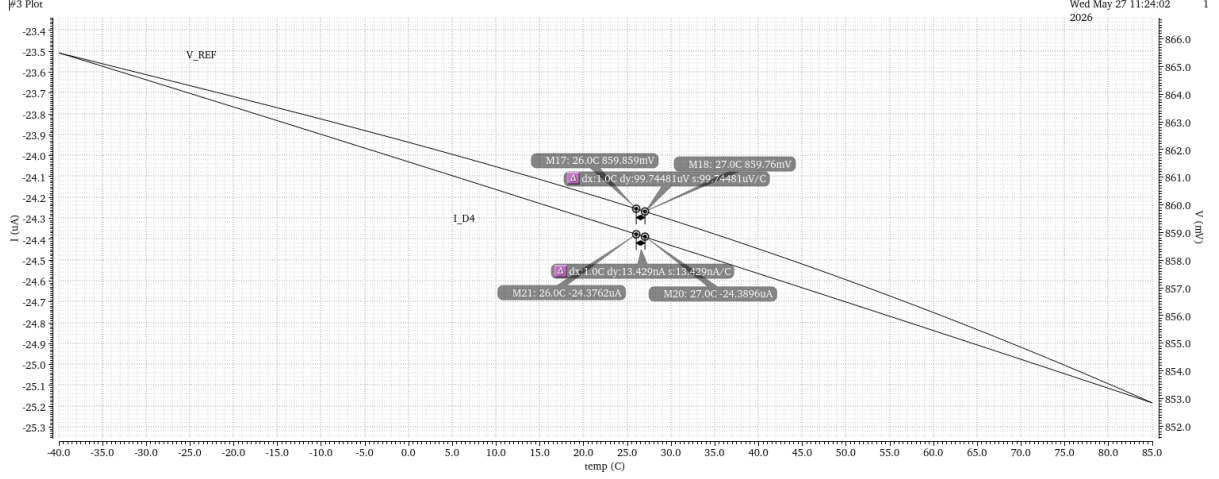


Figure 43: Voltages V_{N1} V_{P1} and the resistance R_1

As for the reference voltage V_{REF} , Fig. 43 shows how this is close to the desired value but still affected by temperature

$$V_{REF} \approx 859.76 \text{ mV}$$

$$\frac{\partial V_{REF}}{\partial T} = 99.7 \frac{\mu V}{C} \quad TC_{VREF} \approx 115.96 \text{ ppm/C}$$

and the behavior of the drain current of transistor $M4$, giving a slope of

$$\frac{\partial I_{D4}}{\partial T} = 13.429 \frac{nA}{C} \quad TC_{Id4} \approx 550.6 \text{ ppm/C}$$

4.3 Resistance Calibration

4.3.1 Calibration of R_1

Choosing initially $R_1 = 3.584 \text{ k}\Omega$ gives in the simulation a diode current $I_{D1} = 11.8997 \mu A$. In order to have it almost equal to $10 \mu A$, the value of the resistance was increased using the following relation :

$$R_1^{new} = R_1^{old} \cdot \left(\frac{I_{sim}}{I_{target}} \right)$$

After some calculation, a value of $R_1 = 4.277978 \text{ k}\Omega$ gives $I_{D1} = 10.0005 \mu A$

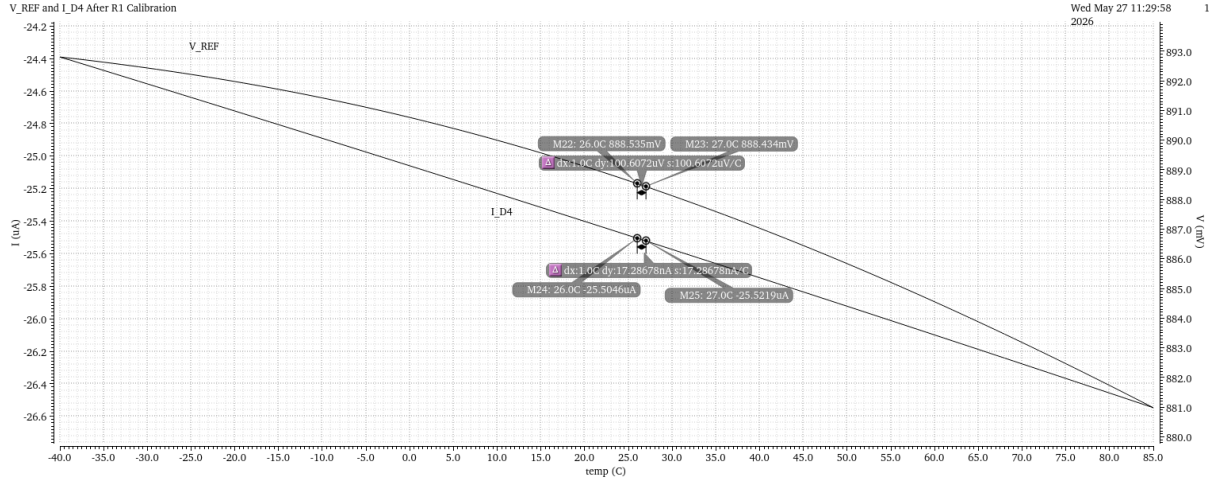


Figure 44: V_{REF} and I_D after changing R_1

After calibrating R_1 (see Fig. 44) :

$$\frac{\partial V_{REF}}{\partial T} = 100 \frac{\mu V}{C} \quad TC_{V_{REF}} \approx 112.56 \text{ ppm/C}$$

$$\frac{\partial I_{D4}}{\partial T} = 17.28678 \frac{nA}{C} \quad TC_{I_{D4}} \approx 677.33 \text{ ppm/C}$$

4.3.2 Calibration of R_2

For the calibration of R_2 a sweep was made in order to find the best values where the slope near the interested range was minimum (see Fig. 45)

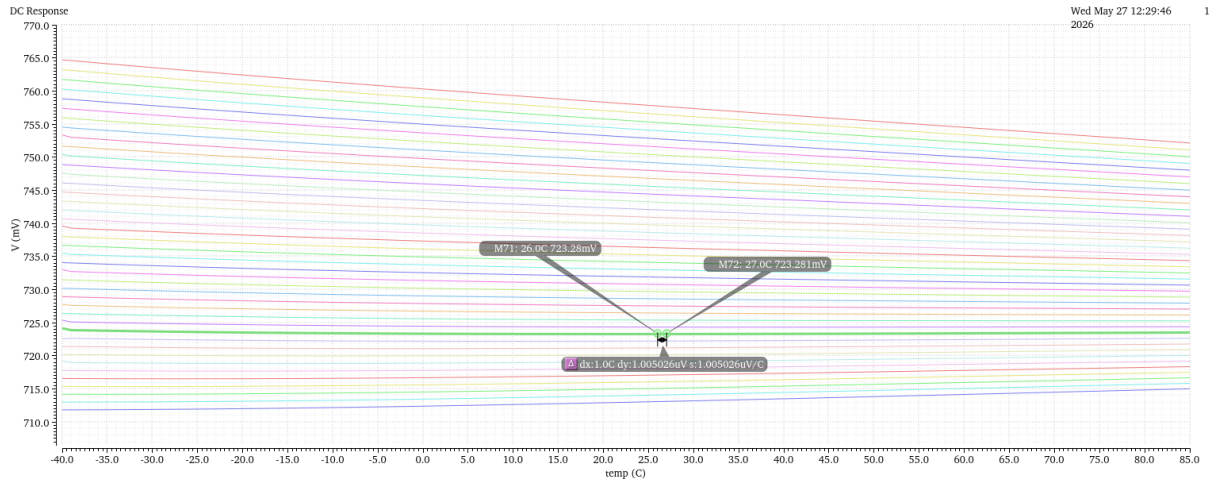


Figure 45: Sweep of R_2

After it, the one with the minimum slope was

$$R_2 = 66 \text{ k}\Omega$$

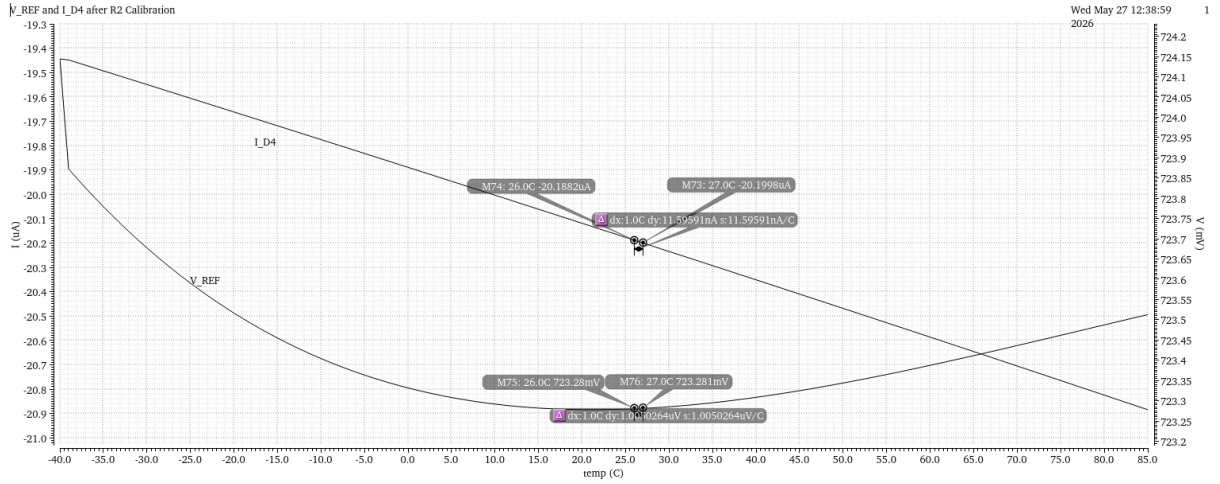


Figure 46: V_{REF} and I_D after changing R_2

After calibrating R_2 (see Fig. 46) :

$$\frac{\partial V_{REF}}{\partial T} = 1 \frac{\mu V}{C} \quad TC_{VREF} \approx 1.38 \text{ ppm/C}$$

$$\frac{\partial I_{D4}}{\partial T} = 11.59 \frac{nA}{C} \quad TC_{ID4} \approx 573.77 \text{ ppm/C}$$

Tuning R_2 balanced the CTAT and PTAT components near the desired temperature (27°C)

4.3.3 Calibration of R_3

To find the new value for the last resistance, the following equation was used :

$$R_{3,new} = \frac{V_{REF}^{ideal}}{I_{R3}^{sim}}$$

After some calculations, the new value of the resistance was found

$$R_3 = 39.8704 \text{ k}\Omega$$

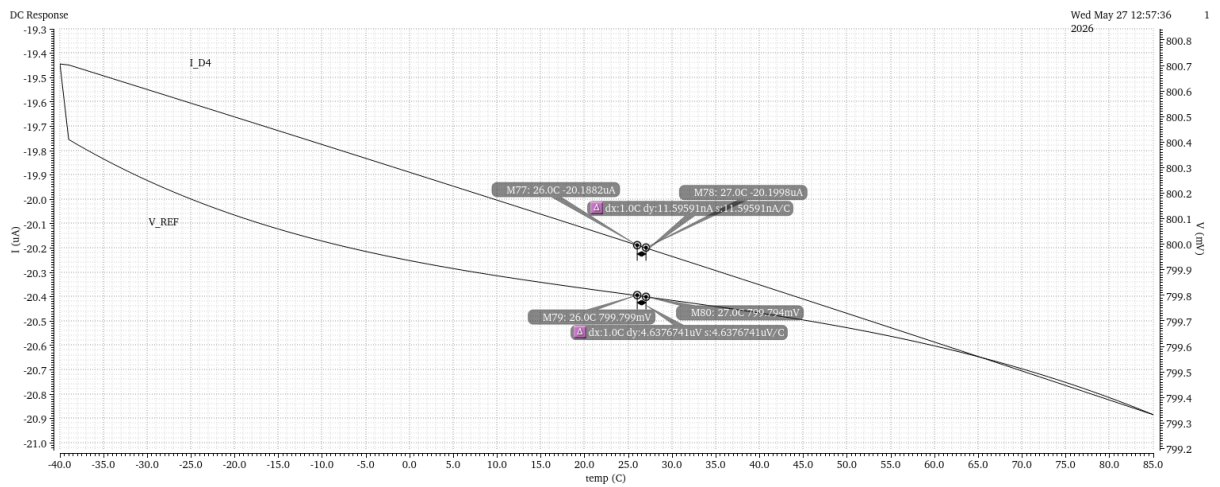


Figure 47: V_{REF} and I_D after changing R_3

After calibrating R_3 (see Fig. 47) :

$$\frac{\partial V_{REF}}{\partial T} = 4.63 \frac{\mu V}{C} \quad TC_{V_{REF}} \approx 5.79 \text{ ppm}/C$$

$$\frac{\partial I_{D4}}{\partial T} = 11.59 \frac{nA}{C} \quad TC_{I_{D4}} \approx 573.77 \text{ ppm}/C$$

In conclusion, the first-order low-voltage Bandgap Reference (BGR) circuit was successfully optimized to balance its complementary-to-absolute-temperature (CTAT) and proportional-to-absolute-temperature (PTAT) components at a nominal room temperature of $27^\circ C$.

The reference voltage was fine-tuned to 799.794 mV, matching the 800 mV design specification; at this operating point, the circuit achieved an exceptional temperature coefficient ($TC_{V_{REF}}$) of approximately 5.79 ppm/ $^\circ C$, driven by a minimized absolute voltage derivative of $4.63 \mu V/^\circ C$.

Concurrently, the current mirror branch demonstrated an expected linear PTAT behavior, characterized by current derivative of $11.59 \text{ nA}/^\circ C$ ($TC_{I_{D4}} \approx 573.77 \text{ ppm}/^\circ C$).

4.4 V_{DD} Sensitivity

To evaluate the sensitivity of V_{DD} , it was performed a V_{DD} sweep around its nominal value, from 1 V to 1.4 V.

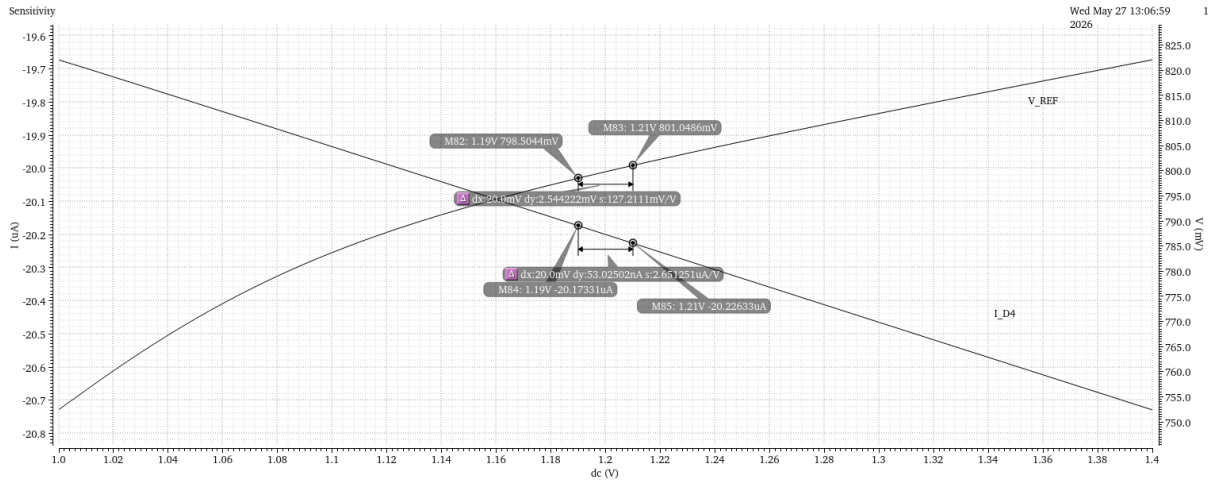


Figure 48: Sensitivity

Looking at Fig. 48 ,thanks to the use of markers around the nominal supply voltage, the sensitivity was estimated as

$$\frac{\partial V_{REF}}{\partial V_{DD}} \approx \frac{V_{REF}(1.21) - V_{REF}(1.19)}{20 \text{ mV}} \approx 127.2 \text{ mV}/V$$

and, similarly, the p-type current mirror sensitivity is

$$\frac{\partial I_{D4}}{\partial V_{DD}} \approx \frac{I_{D4}(1.21) - I_{D4}(1.19)}{20 \text{ mV}} \approx 2.65 \mu V/V$$

The circuit demonstrates robust internal line regulation, as evidenced by the low supply sensitivity of the PMOS current mirror ($\approx 2.65 \mu\text{A}/\text{V}$). However, the final reference voltage remains highly susceptible to supply variations, exhibiting a line regulation error of $127.2 \text{ mV}/\text{V}$.